CCBS 4024
Investments and Portfolio Management

Topic 3: Portfolio theories



Diagrammatic representation of assets' expected returns and risks

- Recall in Lecture 2, given

i. the correlation coefficient (or covariance) between assets,
ii. the expected return of each asset,
iii. the standard deviation of returns of each asset, and
iv. the weight of each asset

We are able to find the **expected return** and **standard deviation** of a portfolio.

- If we **change the weight** of each asset, we have **different portfolio expected returns and standard deviation of returns.**

Summary of the example in Lecture 2

Year	Stock X	Stock Y
2018	18%	0%
2019	5%	-3%
2020	12%	15%
2021	4%	12%
2022	6%	1%

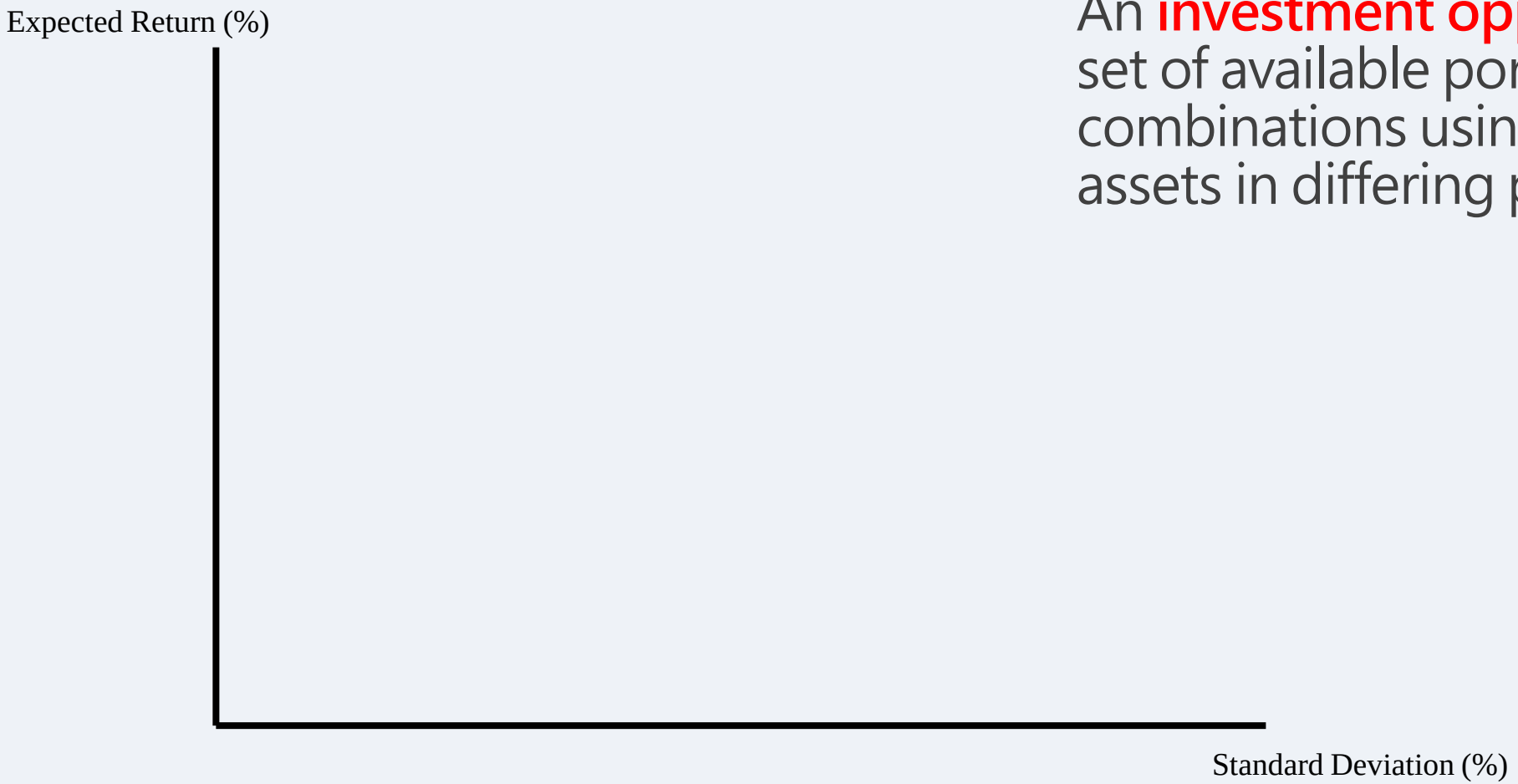
- Mean return of Stock X (Proxy of expected return of X) = 0.09
- Mean return of Stock Y (Proxy of expected return of Y) = 0.05
- Sample standard deviation of return of Stock X = 0.0591608
- Sample standard deviation of return of Stock Y = 0.0796869
- Sample covariance between returns of Stock X and Y = - 0.00015
- Correlation coefficient of Stock X and Stock Y = - 0.031817817

Portfolio expected returns and S.D. with different asset weights

Portfolio	A	B	C	D	E
Weight of Stock X, w_X	100%	60%	40%	20%	0%
Weight of Stock Y, w_Y					
$E(R_p)$					
σ_p					

Calculate the expected return and S.D. of different portfolios above. Then represent them in an expected return-S.D. space in the next slide.

Investment opportunity set of Stock X and Y



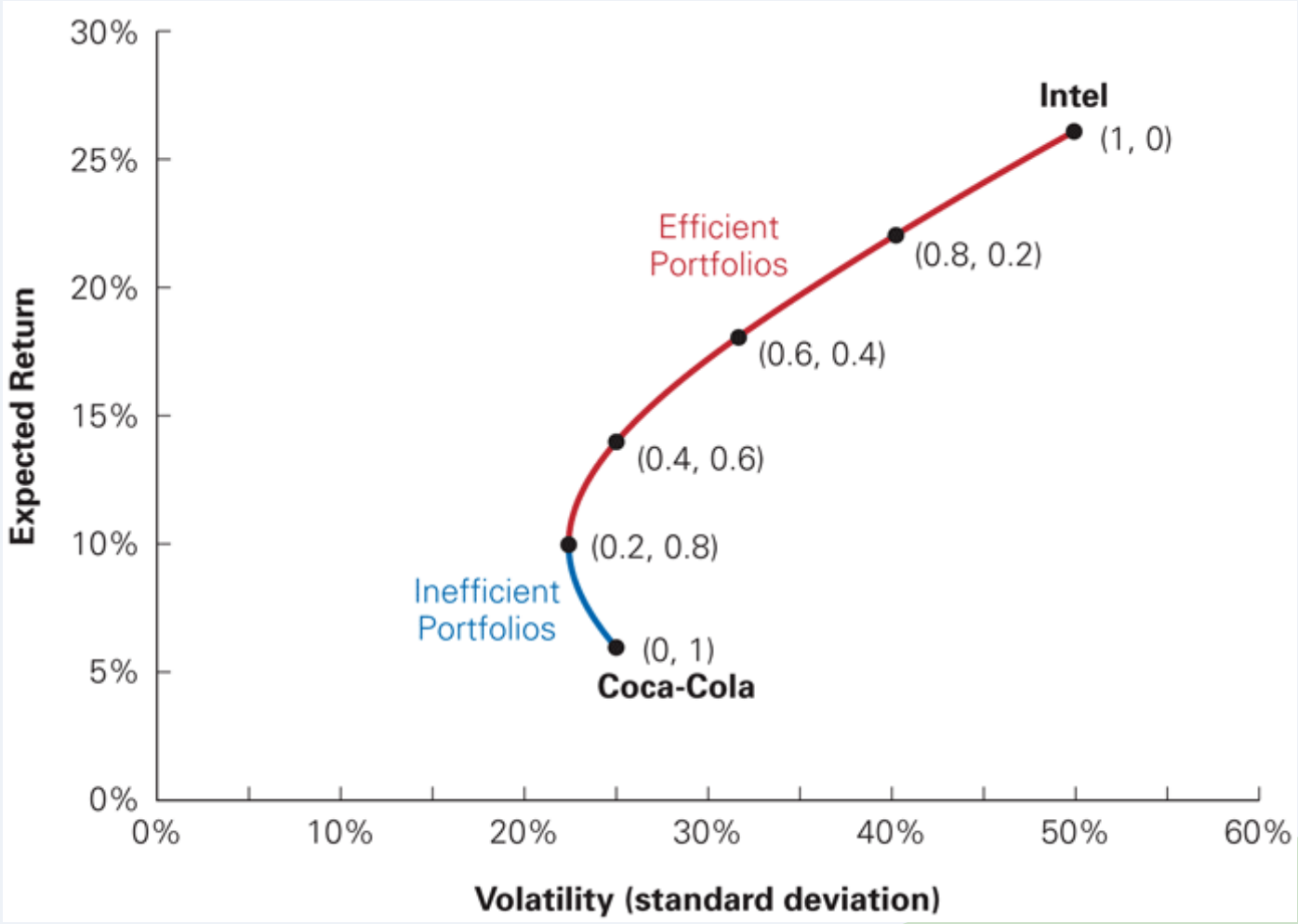
An **investment opportunity set** is the set of available portfolio risk-return combinations using the available assets in differing proportions.

Example: Investment Opportunity Set

Consider Expected Returns and Volatility for Different Portfolios of Two Stocks, Intel and Coca-Cola as follows:

Portfolio Weights		Expected Return (%)	Volatility (%)
x_I	x_C	$E[R_P]$	$SD[R_P]$
1.00	0.00	26.0	50.0
0.80	0.20	22.0	40.3
0.60	0.40	18.0	31.6
0.40	0.60	14.0	25.0
0.20	0.80	10.0	22.4
0.00	1.00	6.0	25.0

The investment opportunity set of portfolios consisting Intel and Coca-Cola



Mean-variance criterion



If $E(r_A) \geq E(r_B)$ and $\sigma_A \leq \sigma_B$, Portfolio A dominates Portfolio B.



Investors prefer portfolios with higher mean returns and lower volatility.

Exercise

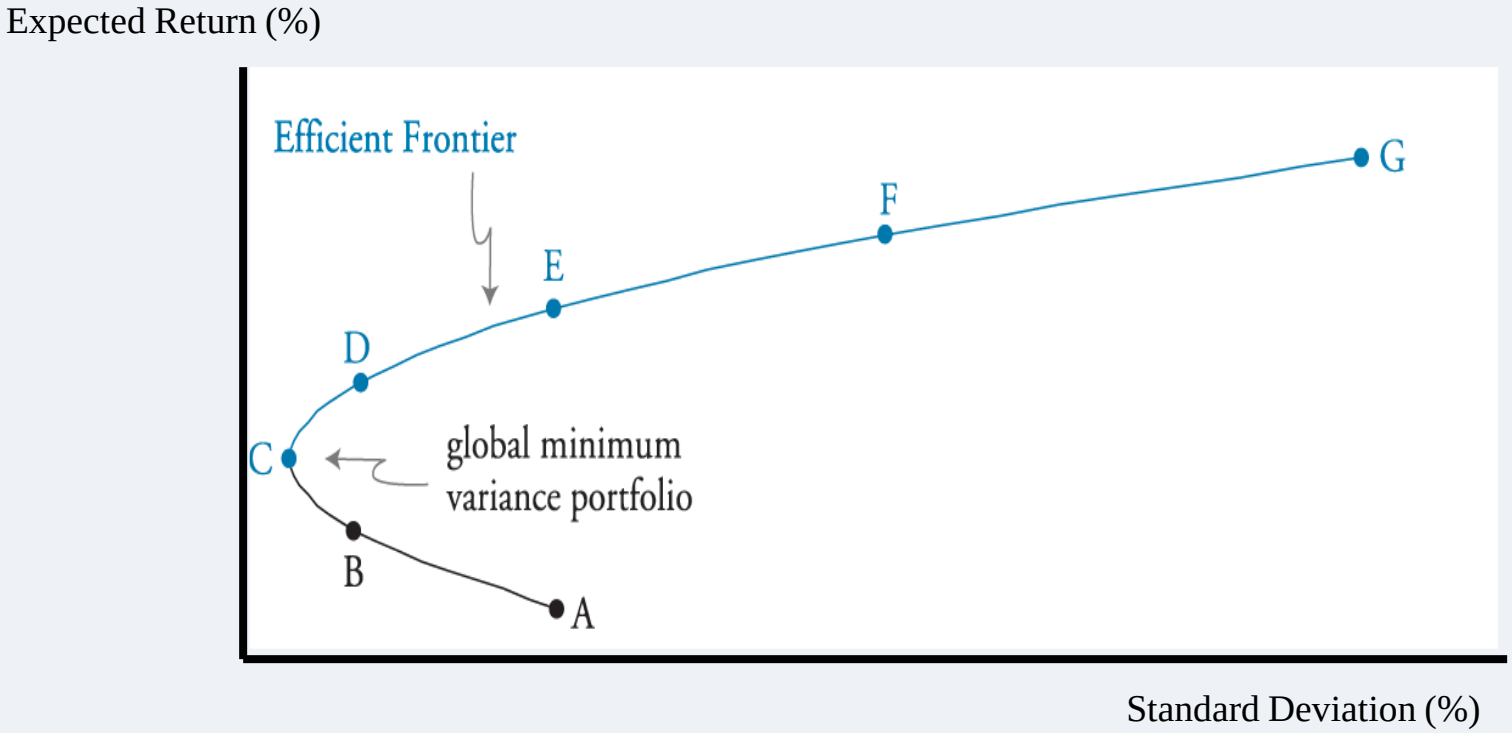
For slide 6, Sally has invested 100% of her money in coca-cola stock and is seeking investment advice. She would like to earn the highest expected return possible without increasing her S.D. Which portfolio would you recommend?

Answer

In the diagram on slide 7, we can see that Sally can invest up to 40% in Intel stock without increasing her volatility. Because Intel stock has a higher expected return than coca-cola stock, she will earn higher expected returns by putting more money in Intel stock.

Therefore, you should recommend Sally to put 40% of her money in Intel stock, leaving 60% in coca-cola stock. This portfolio has the same volatility of 25%, but an expected return of 14% rather than the 6% she has now.

Minimum variance frontier, global minimum variance portfolio and the efficient frontier



Minimum variance frontier, minimum variance portfolio and the efficient frontier

- The above graph is also called **the minimum-variance frontier** which minimizes portfolio's S.D. given its expected returns.
- On a typical minimum-variance frontier, the portfolio with the smallest variance or standard deviation among all possible portfolios is called **global minimum-variance portfolio**.
- **Efficient portfolios** mean there is no way to reduce the S.D. of the portfolio without lowering its expected return. Or the expected return of portfolio is maximized given the same S.D.
- The **efficient frontier** shows all efficient portfolios, all of which lie along the upper portion of the minimum-variance frontier.

Two-risky-asset Investment Opportunity Set with the correlation coefficient between 1 and -1

- If the correlation between the two assets is between 1 and – 1, the investment opportunity set is a curve joining two assets and bending to left.
- If we lower the correlation coefficient of two assets, the diversification benefits will be (smaller / larger). Thus, given the same portfolio expected return, the portfolio standard deviation will be (lower / higher). The investment opportunity set will bend to the left to a (smaller / greater) degree.

Two-risky-asset's Investment Opportunity Set: extreme cases

- If the correlation between the two assets is +1, the investment opportunity set is a **straight line** joining two assets.
- If the correlation between the two assets is **–** 1, the investment opportunity set consists **of two straight line segments** meeting at a "V" or sharp angle on the vertical expected return axis.



Investment Opportunity Set with different correlation between 2 assets

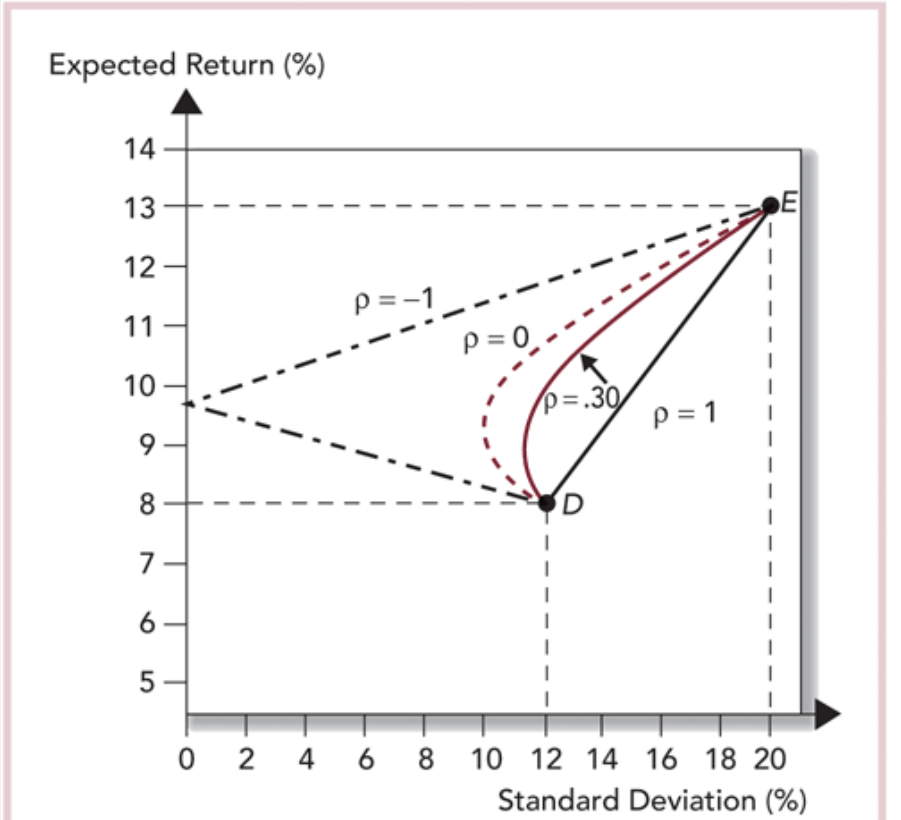


FIGURE 7.5 Portfolio expected return as a function of standard deviation

A portfolio with three risky assets

- Suppose we form a portfolio using **three risky assets, A, B and C**

➢ The expected return on three-asset portfolio = $E(R_p) = w_A E(r_A) + w_B E(r_B) + w_C E(r_C)$

➢ The three-asset portfolio's variance is

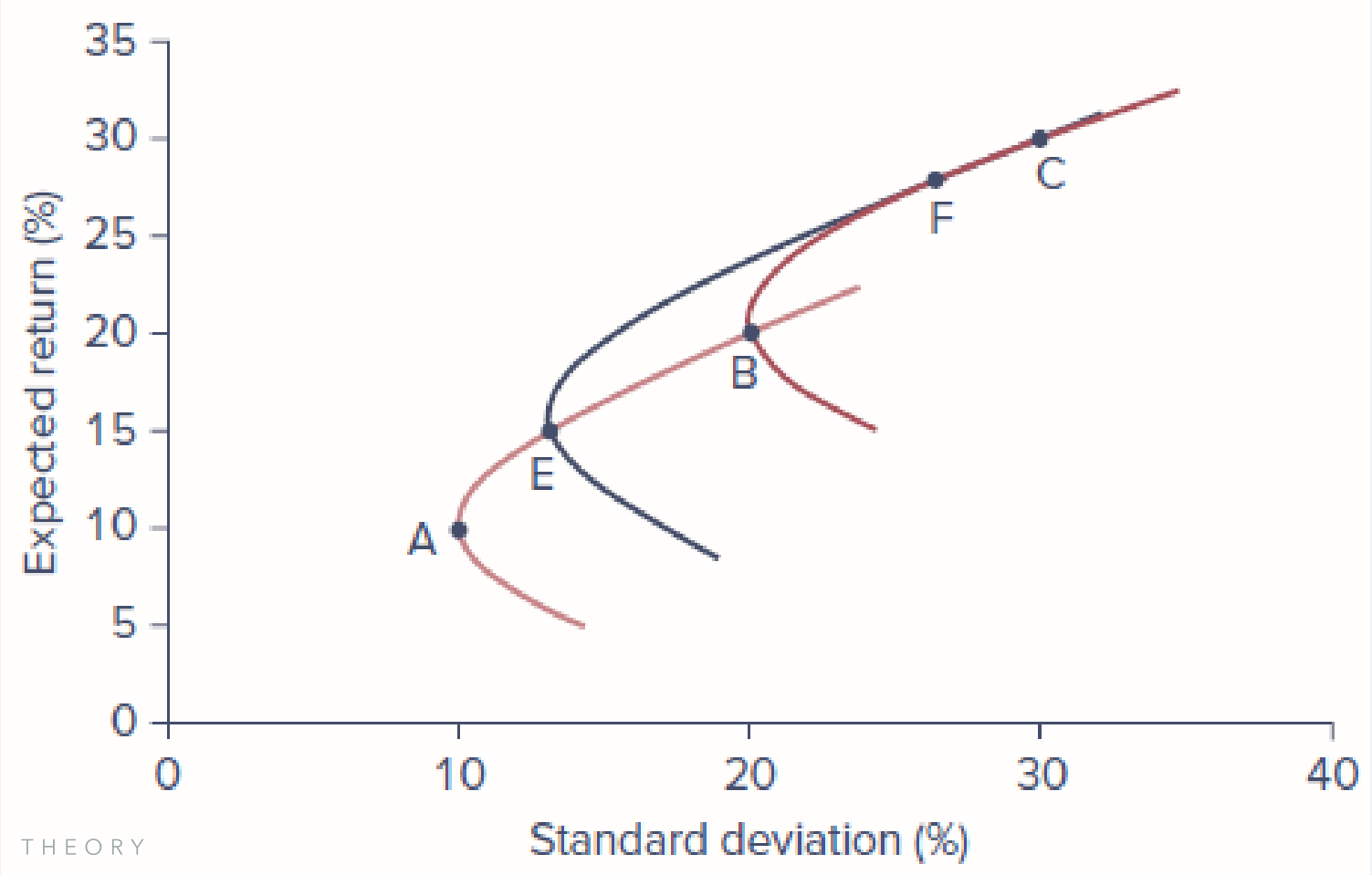
$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + w_C^2\sigma_C^2 + 2w_A w_B Cov(r_A, r_B) + 2w_A w_C Cov(r_A, r_C) + 2w_B w_C Cov(r_B, r_C)$$

➢ The opportunity set of three-risky-asset portfolio is an **area** in expected return-S.D. space.

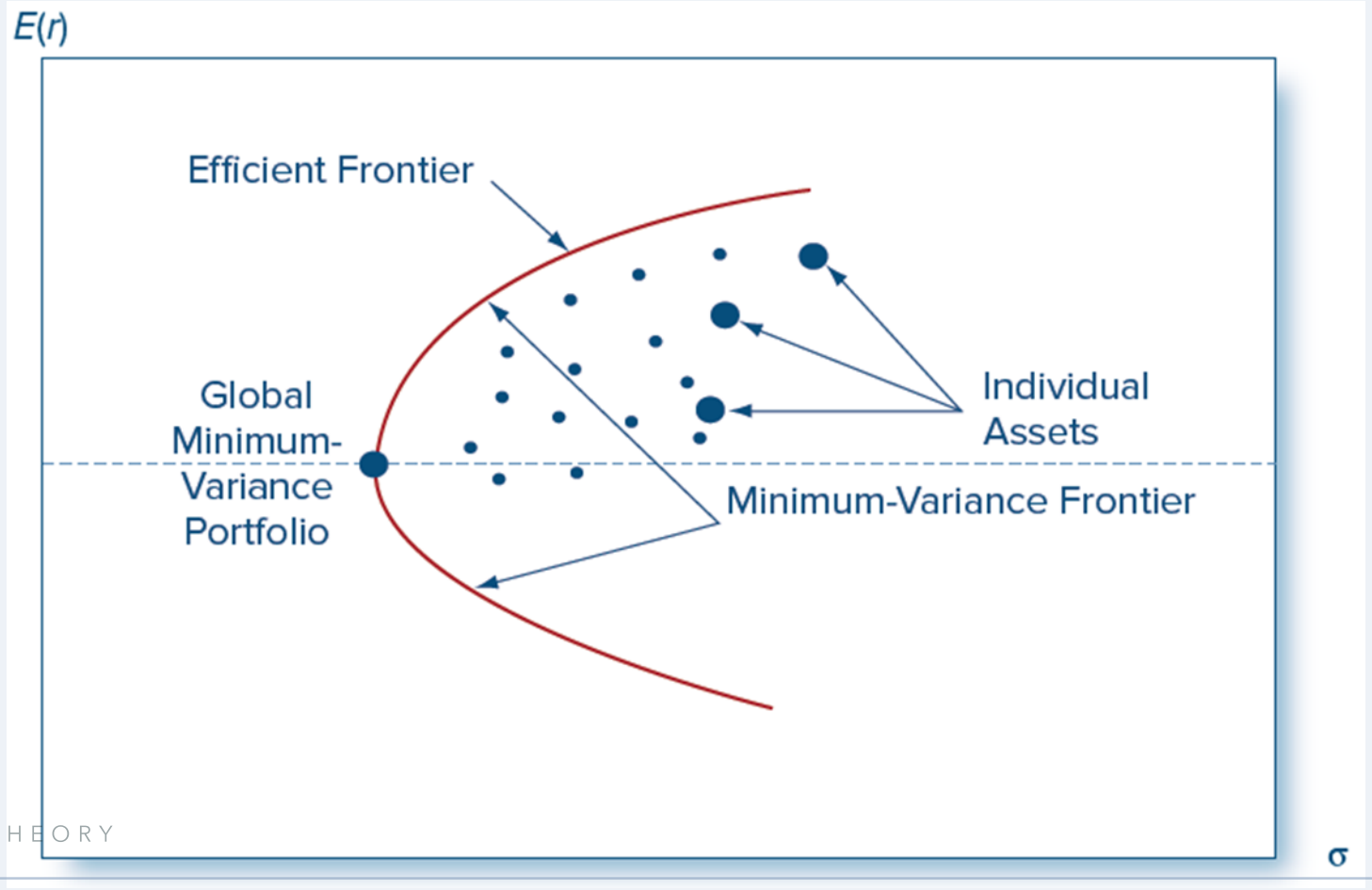
➢ From the opportunity set, we can find the minimum variance frontier, global minimum variance portfolio, the efficient frontier similar to the case of two-risky-asset portfolios.

➢ Similar technique can also be applied to portfolios with more than 3 risky assets.

Portfolios Constructed with Three Stocks



Minimum variance frontier with many risky assets



Optimal Risky Portfolio

- Given the minimum variance frontier, which portfolio will be chosen by an investor? What is the investor's **optimal risky portfolio**?
- The optimal risky portfolio depends on how investors perceive risk and expected return of the risky portfolios (risk preference of investors).
- Investors are better off if assets offer **higher expected returns** (good) but lower **risk** (bad). Investors who dislike risk is called **risk-averse** investors.
- The risk preference of investors can be represented by the following ways:

1. Diagrams (Indifference Curves)
2. Mathematical functions (Utility functions)

Indifference curves

- **Indifference curves** show different combination of expected returns and standard deviation of returns that investors are indifferent.

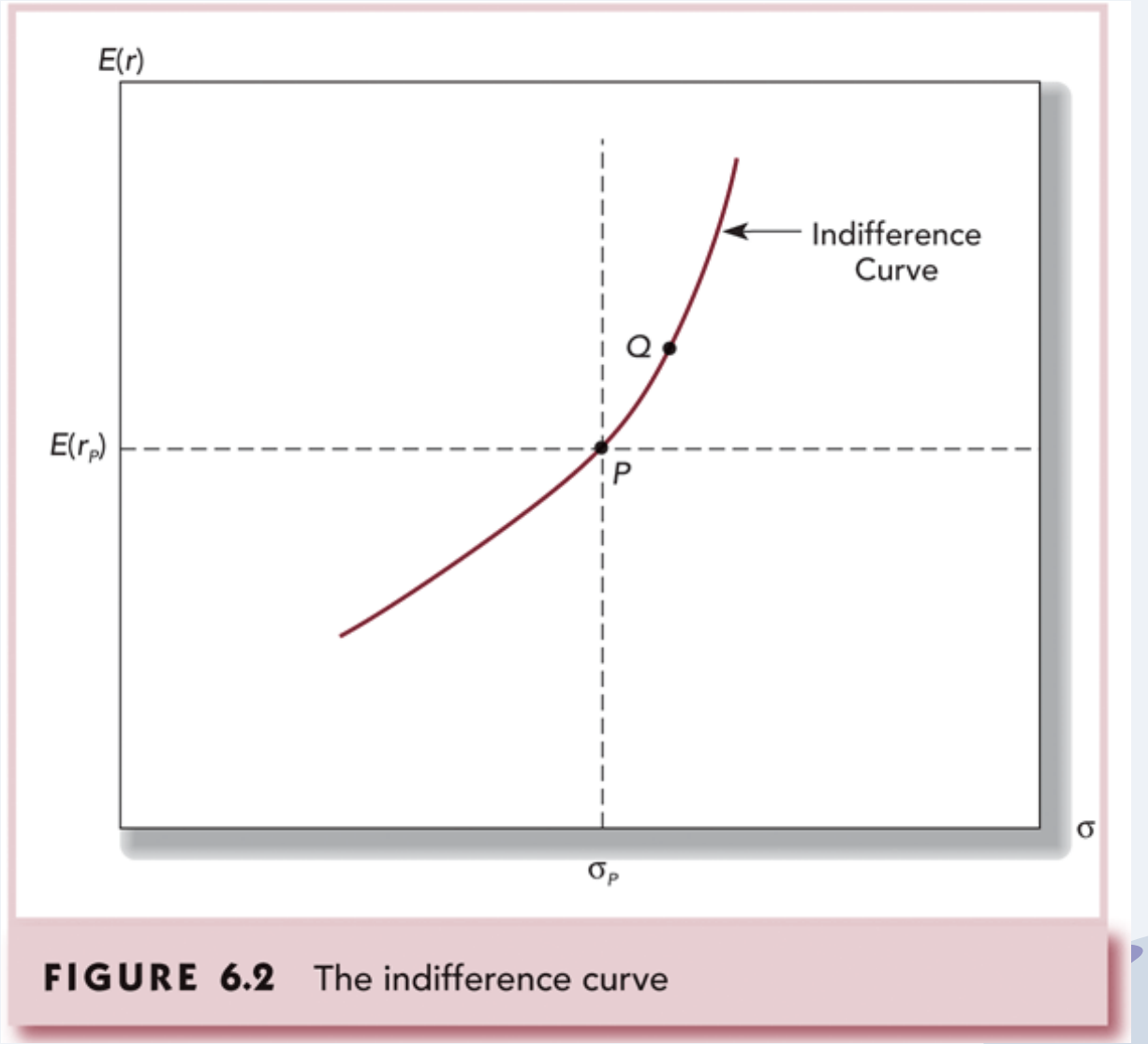
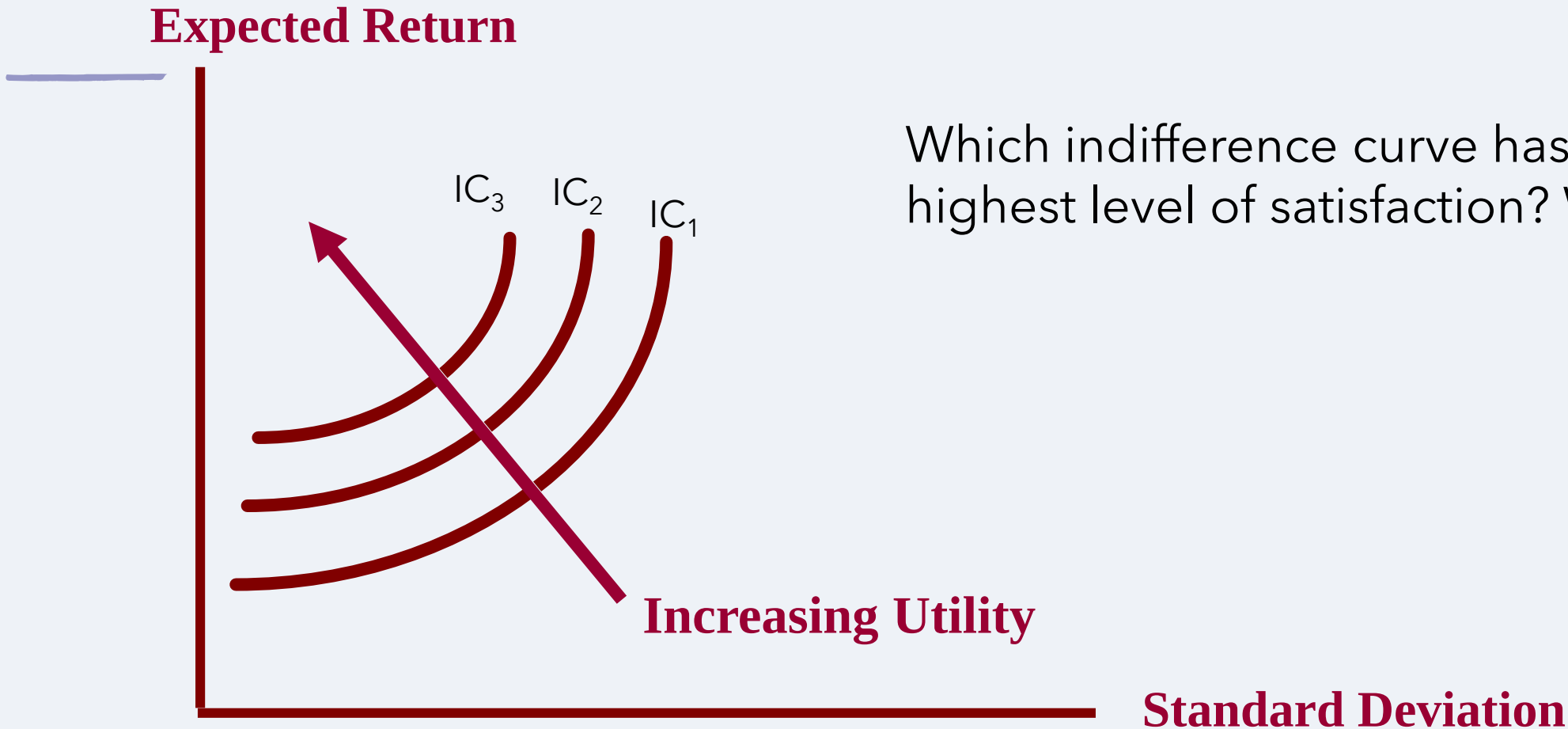


FIGURE 6.2 The indifference curve

Indifference Curves



Which indifference curve has the highest level of satisfaction? Why?

Properties of Indifference curves

1. **Risk averse investors** have indifference curves with a positive slope, implying that risk and return are positively related. In order to maintain equal satisfaction, the risk-averse investor demands a higher expected return in exchange for higher risk.
2. A higher or more preferred indifference curve lies in the northwest direction.
3. Indifference curves of an investor never cross.
4. The steeper the IC, the more risk averse is the investor.

Utility function

The mathematical counterpart of IC is called **utility function**.

For example, $Utility\ (U) = E(\mathrm{r}) - \frac{1}{2}\mathrm{A}\sigma^2$ where

U is the utility value;

E(r) is the expected return of the portfolio;

A is an index of the investor's risk aversion; and

σ^2 is the variance of the portfolio's return.

- We can calculate the utility value of different portfolios and investors will choose the portfolio with the highest utility value.
- The utility function shows that utility increases with expected return but falls with volatility.

Example

Use the following utility function to determine which portfolio dominates the other.

$Utility\ (U) = E(\mathrm{r}) - \frac{1}{2}\mathrm{A}\sigma^2$ where A equals 4.

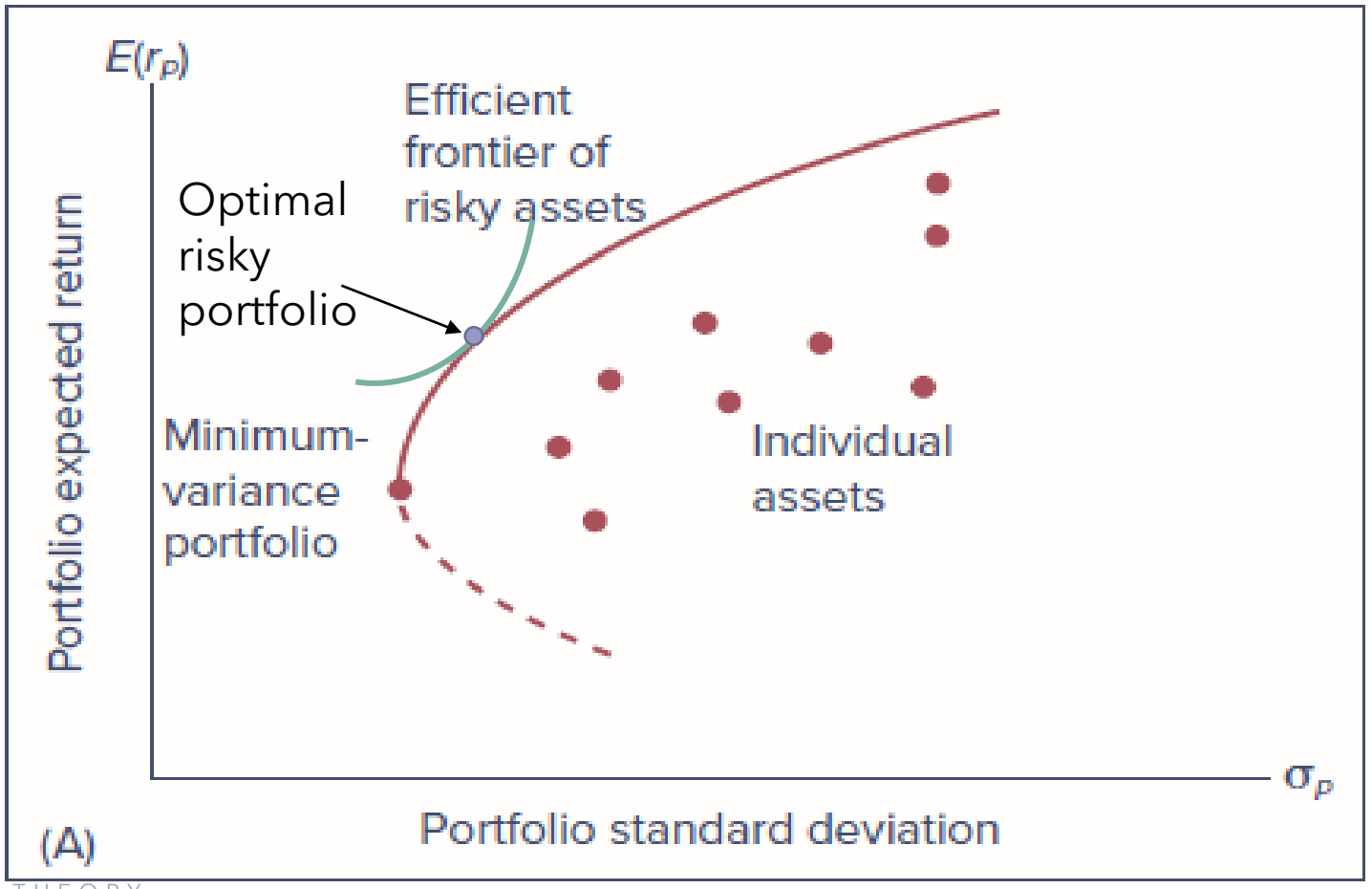
Portfolio	Expected Return	S.D.
X	10%	8%
Y	6%	5%

Optimal risky portfolio

Given the efficient frontier constructed from a lot of risky assets, an investor would choose the utility maximizing portfolio or optimal risky portfolio **at the tangency point of IC and efficient frontier**. Refer to the next slide.

If the investor is more risk-averse, the IC is (flatter / steeper), and the investor will choose the optimal risky portfolio with (lower / higher) S.D. but (lower / higher) expected return than that of less risk-averse investors.

Optimal risky portfolio



Panel A

Key:
Indifference Curve

A portfolio with one risky asset and the risk-free asset

- Suppose we form a portfolio using one risky asset, A and a risk-free asset, r_f (instead of two risky assets).

➢ The expected return on the portfolio = $w_A E(r_A) + (1 - w_A)(r_f)$

➢ The portfolio's standard deviation = $\sigma_p = w_A \sigma_A$

- By combining the above two equations, the feasible set portfolio with the two assets (a risky asset and the risk-free asset) is called **Capital Allocation Line (CAL)**:

$$\mathrm{E}(r_p) = r_f + \left[\frac{E(r_A) - r_f}{\sigma_A}\right]\sigma_p$$

Capital Allocation Line (CAL)

- The equation of Capital Allocation Line (CAL) is:

$$\mathrm{E}(r_p) = r_f + \left[\frac{E(r_A) - r_f}{\sigma_A}\right]\sigma_p$$

Features of CAL (A):

1. It is a linear straight line joining the risk-free asset and Asset A.
2. Its y-intercept is r_f.
3. Its slope is $\frac{E(r_A)-r_f}{\sigma_A}$ which is also called **Sharpe ratio**.

Example of CAL

Suppose

$\mathbf{r_f = 7\%}$	$\sigma_{rf} = \mathbf{0\%}$
$\mathbf{E(r_A) = 15\%}$	$\sigma_A = \mathbf{22\%}$

Equation of CAL(A): $\mathrm{E}(r_p) = r_f + \left[\frac{E(r_A) - r_f}{\sigma_A}\right]\sigma_p$

$$\mathrm{E}(r_p) = 7\% + \left(\frac{15\% - 7\%}{22\%}\right)\sigma_p \text{ or } \mathrm{E}(r_p) = 0.07 + 0.36\sigma_p$$

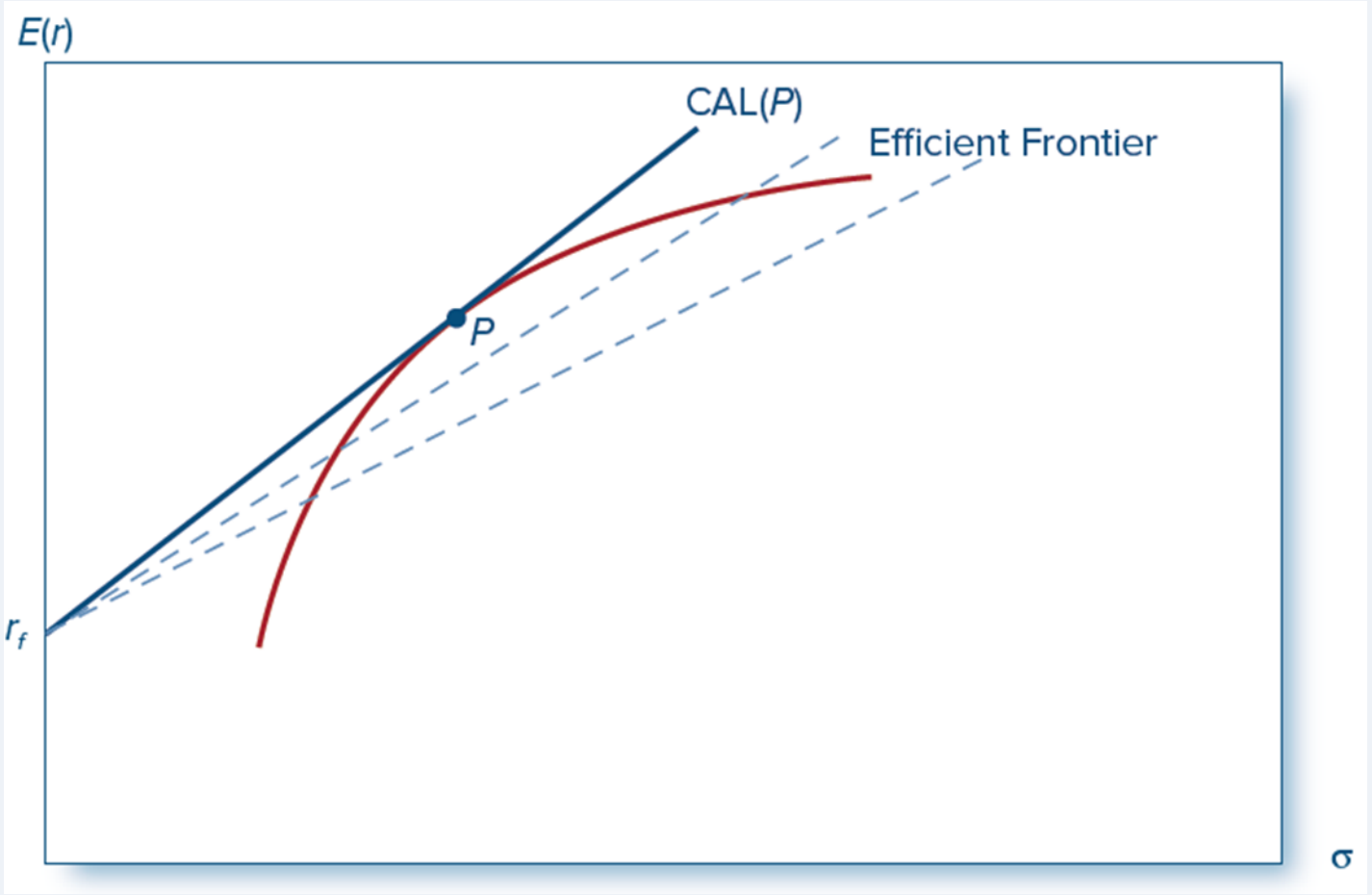
Capital Allocation Line (CAL (A))



How should the investor choose among the risky portfolios to combine with the risk-free asset?

- In practice, investors can form a CAL with a risky asset/portfolio and risk-free asset.
- With many risky portfolios, there are more than one CALs. Which CAL is the best?
- Investors should choose the CAL that maximizes the **reward-to-risk ratio or Sharpe ratio**. In other words, these portfolios on the best CAL always give investors higher expected returns given the identical standard deviation.
- Graphically, investors only consider the Capital Allocation Line which lies tangent to the efficient frontier.

The best CAL



Markowitz Portfolio Optimization Model

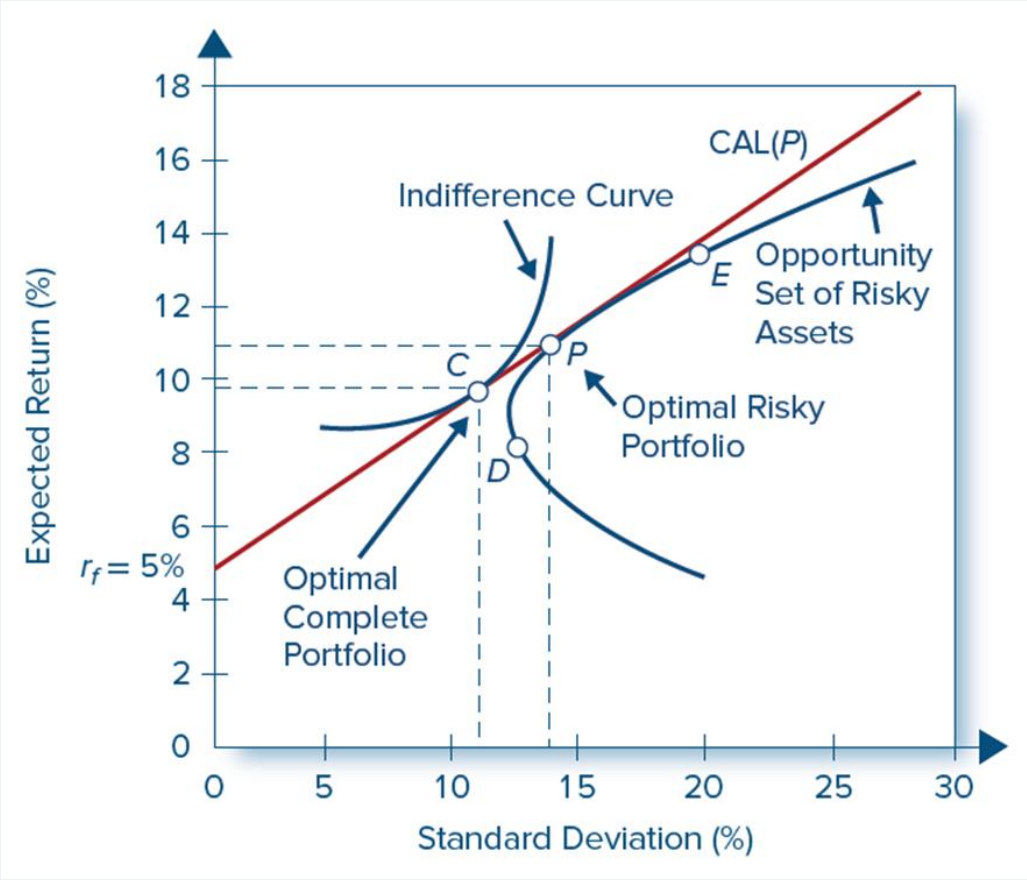
The **separation property:** Portfolio choice problem may be separated into two independent tasks.

1. **Determination of the optimal risky portfolio** is purely technical → the CAL with the highest Sharpe Ratio.
2. **The construction of the optimal complete portfolio** from risk-free assets and the optimal risky portfolio depends on the personal risk preference (IC).

Implications:

All investors (regardless of age and occupation) will hold a portfolio made up of a risk-free asset and the same optimal risky portfolio. They choose different weight on the two assets according to their risk preference.

Optimal Complete Portfolio



Capital Asset Pricing Model (CAPM)

- The CAPM is a set of predictions concerning equilibrium expected returns on risky assets.
- Based on two sets of assumptions.

1. Individual behavior.
2. Market structure.

- Markowitz established modern portfolio management in 1952.
- Sharpe, Lintner and Mossin published CAPM in 1964.

Assumptions of CAPM

1. Individual behavior.
a. Investors are rational, mean–variance optimizers.
b. Their common planning horizon is a single period.
c. Investors all use identical input lists, an assumption often termed **homogeneous expectations.** Homogeneous expectations are consistent with the assumption that all-relevant information is publicly available.
2. Market structure.
a. All assets are publicly held and trade on public exchanges.
b. Investors can borrow or lend at a common risk-free rate, and they can take short positions on traded securities.
c. No taxes.
d. No trading costs.

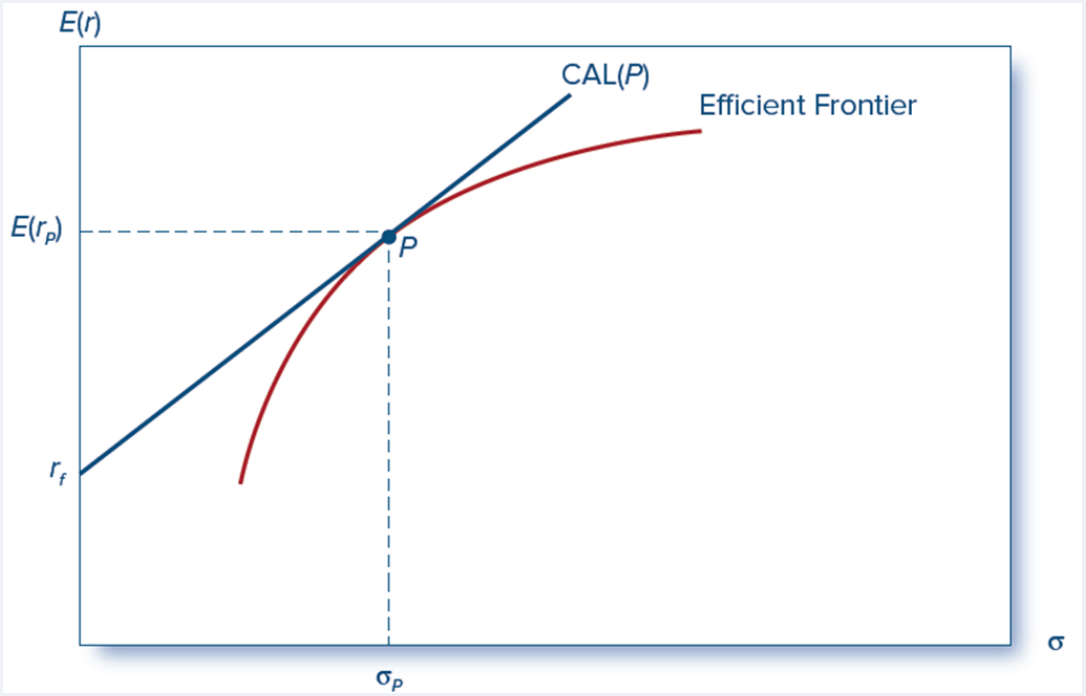
The Market Portfolio

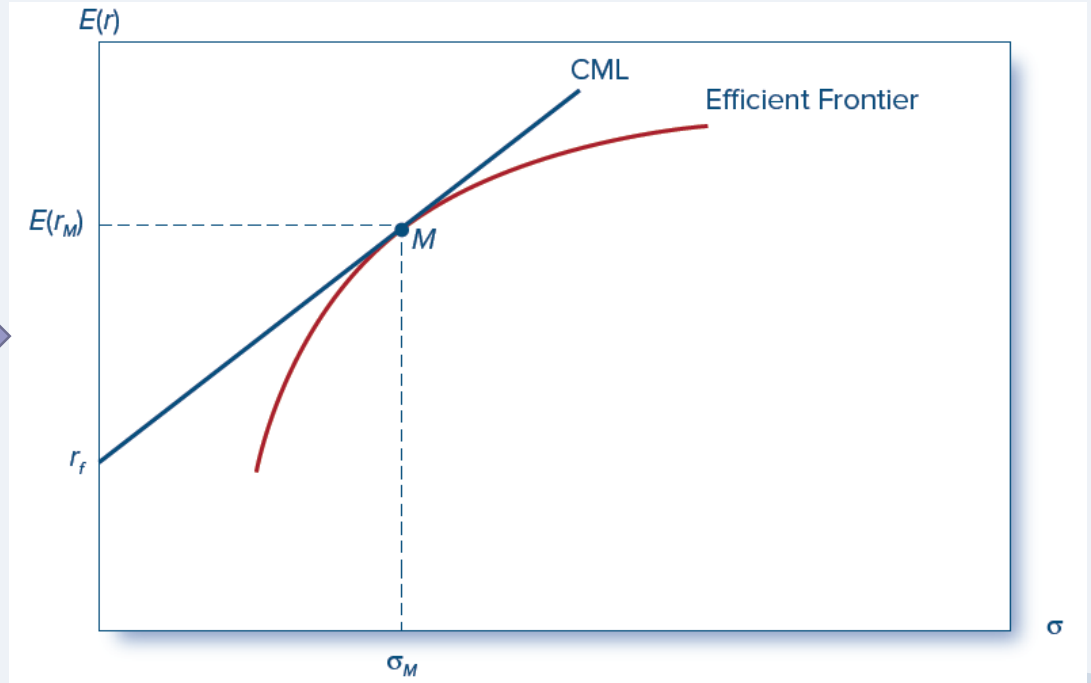
If all assumptions hold in the above slide, all investors will have the same efficient frontier, the same CAL and hold the same portfolio for risky assets (Optimal risky portfolio).

The same optimal risky (tangency) portfolio held by all investors is known as the market portfolio (M).

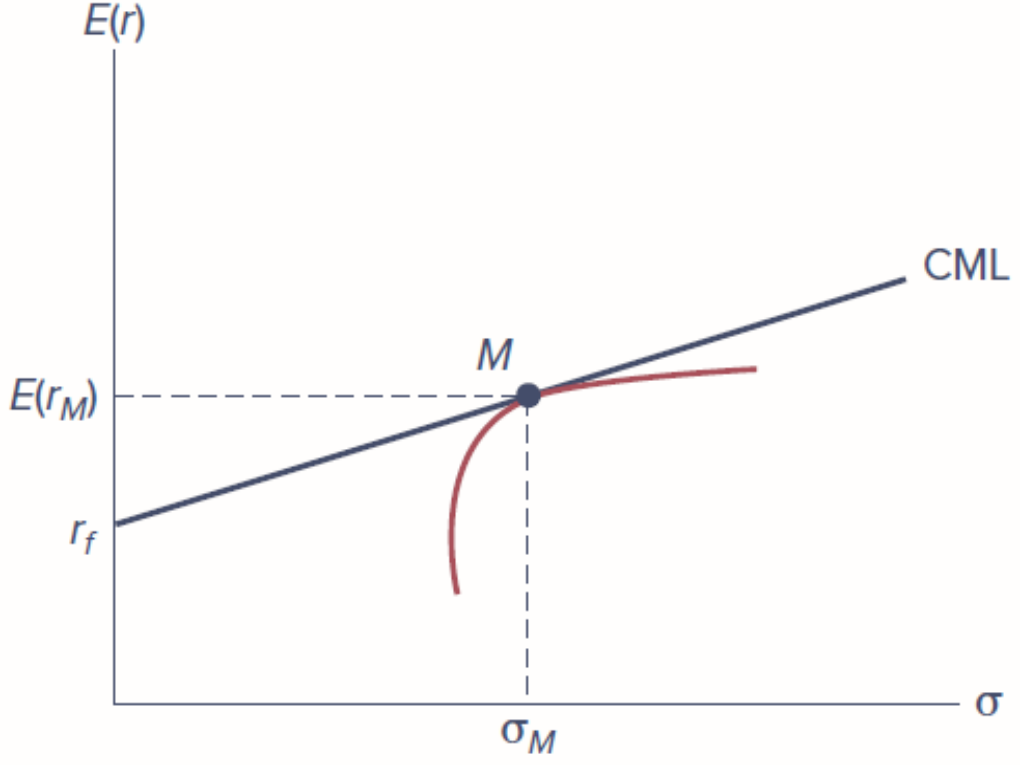
The best CAL is then called the **Capital Market Line (CML).**

The Efficient Frontier and the Capital Market Line





Properties of Capital Market Line (CML)



- Recall CML is also a CAL. We can replace portfolio A by the market portfolio M in CML.
- The equation of CML is:

$$\mathrm{E}(r_p) = r_f + \left[\frac{E(r_M) - r_f}{\sigma_M}\right]\sigma_p$$

Properties of CML:

1. y-intercept: _____
2. Slope: _____
3. $E(r_M) - r_f$ is called market risk premium

Limited use of Capital Market Line (CML)

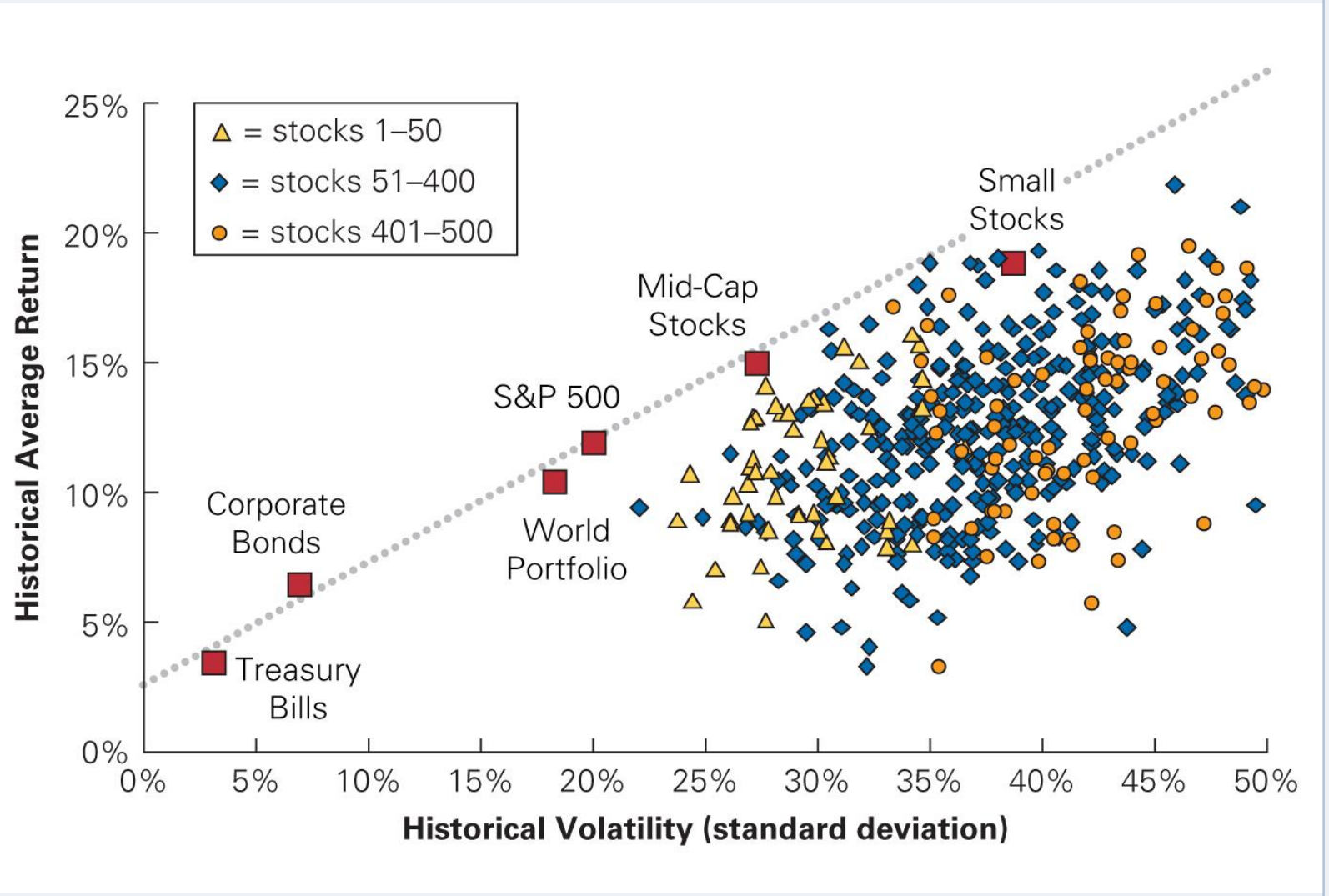
- The CML can find the equilibrium return of those large and diversified portfolios only given their standard deviation. However, it is not quite useful for finding the equilibrium return of **individual assets** as they won't fall on the CML.

- In other words, there is no positive relationship between standard deviation and average returns for individual stocks. It is not true to say higher S.D., higher expected return for an individual security as part of the risk of an individual security can be eliminated easily by diversification.

Limited use of Capital Market Line (CML)

- The market won't reward investors with higher expected returns for bearing more diversifiable risk!!!
- Only when investors bear more non-diversifiable risk, β, they will get higher expected return from their investments at the equilibrium.
- We then need to develop a new tool to link up the relationship between beta (β) of an investment with its equilibrium return.

Historical Volatility and Return for 500 Individual Stocks, Ranked Annually by Size



Capital Asset Pricing Model (CAPM)

- The CAPM provides a way to calculate an asset's expected return ("required" return or equilibrium return) based on its level of systematic (or market-related) risk, as measured by the asset's beta.
- The **Security Market Line (SML)** describes the relationship between the required return and systematic risk for all assets, both individual securities and portfolios.
- This equation of the Security Market Line (SML) is:

$$E(R_i) = R_f + [E(R_M) - R_f]\beta_i$$

Interpretation

The required return of asset i is equal to the sum of risk-free rate and the risk premium of asset i.

Properties of the SML: $E(R_i) = R_f + [E(R_M) - R_f]\beta_i$

1. The required return of an asset can be decomposed into two parts:
   I) risk-free return and
   II) the risk premium of the asset, $\left[\mathrm{E}(R_M) - r_f\right]\beta_i$
2. The relevant risk affecting individual asset's required return is the systematic risk, β_i but not the standard deviation (total risk).
3. The systematic risk, β_i and the expected return is linearly related.
4. The y-intercept of the SML is _____.
   The slope of the SML is _____.

The graph of the Security Market Line (SML)

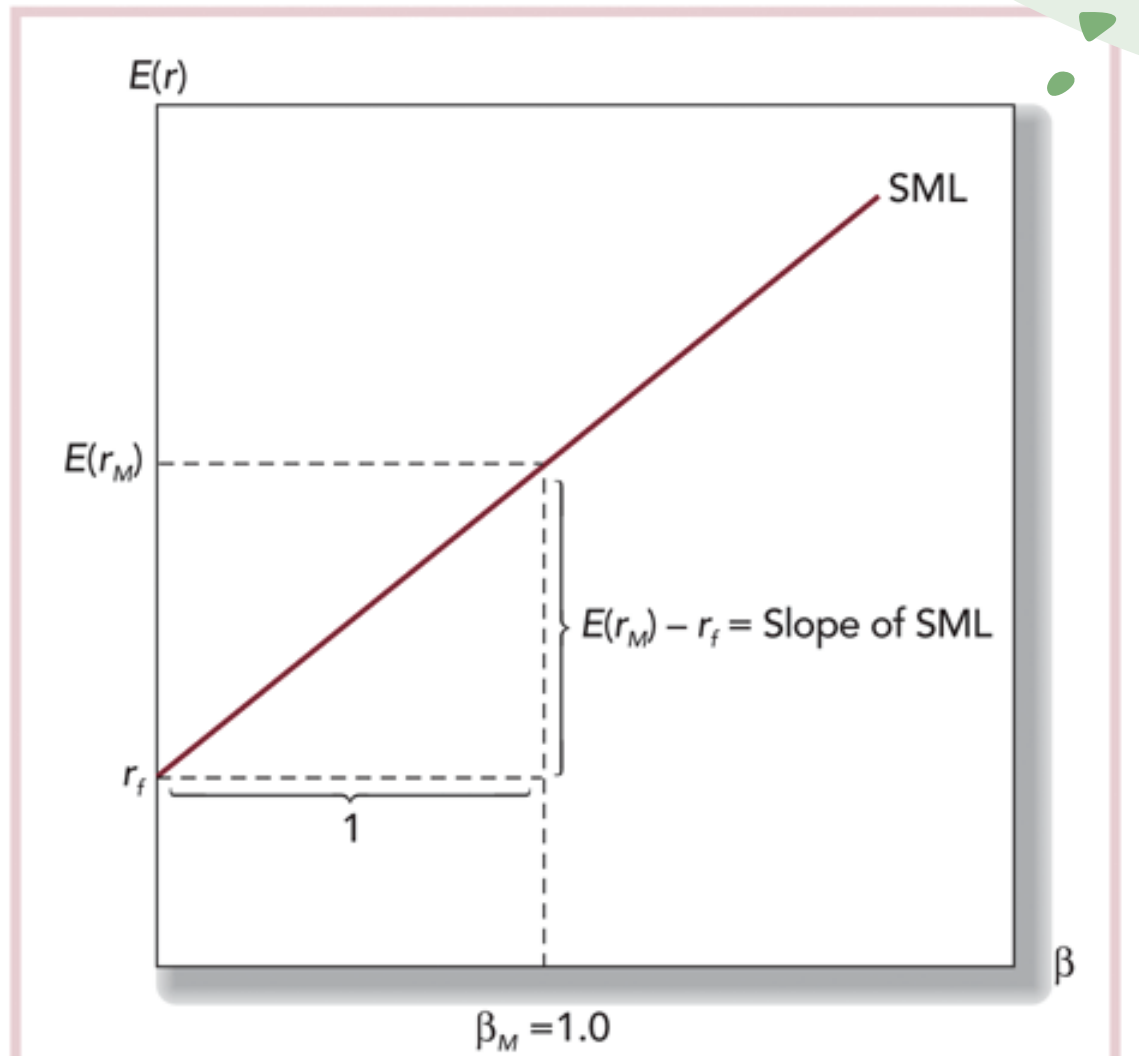


FIGURE 9.2 The security market line

Example of using SML

1. Suppose the risk-free rate is 2%, and the expected market return is 10%. An investor has a portfolio with a beta of 1.5. What is the expected return on the portfolio?
2. If the Security Market Line for a certain industry has a slope of 0.8, what is the beta of a stock in that industry that has an expected return of 15% and a risk-free rate of 3%?
3. What is the required rate of return on a stock with a beta of 1.2 if the risk-free rate is 1% and the expected market return is 9% based on the Security Market Line?
4. If a stock has an expected return of 16% and a beta of 1.8, what is the expected market return according to the Security Market Line if the risk-free rate is 4%?

Uses of SML in the discounted dividend model

Suppose the risk-free return is 4% and the market portfolio has an expected return of 10%. The beta of Stock A is 1.5. If Stock A is expected to pay $2 dividend per share next year and forever, what is the fair stock price of Stock A now?

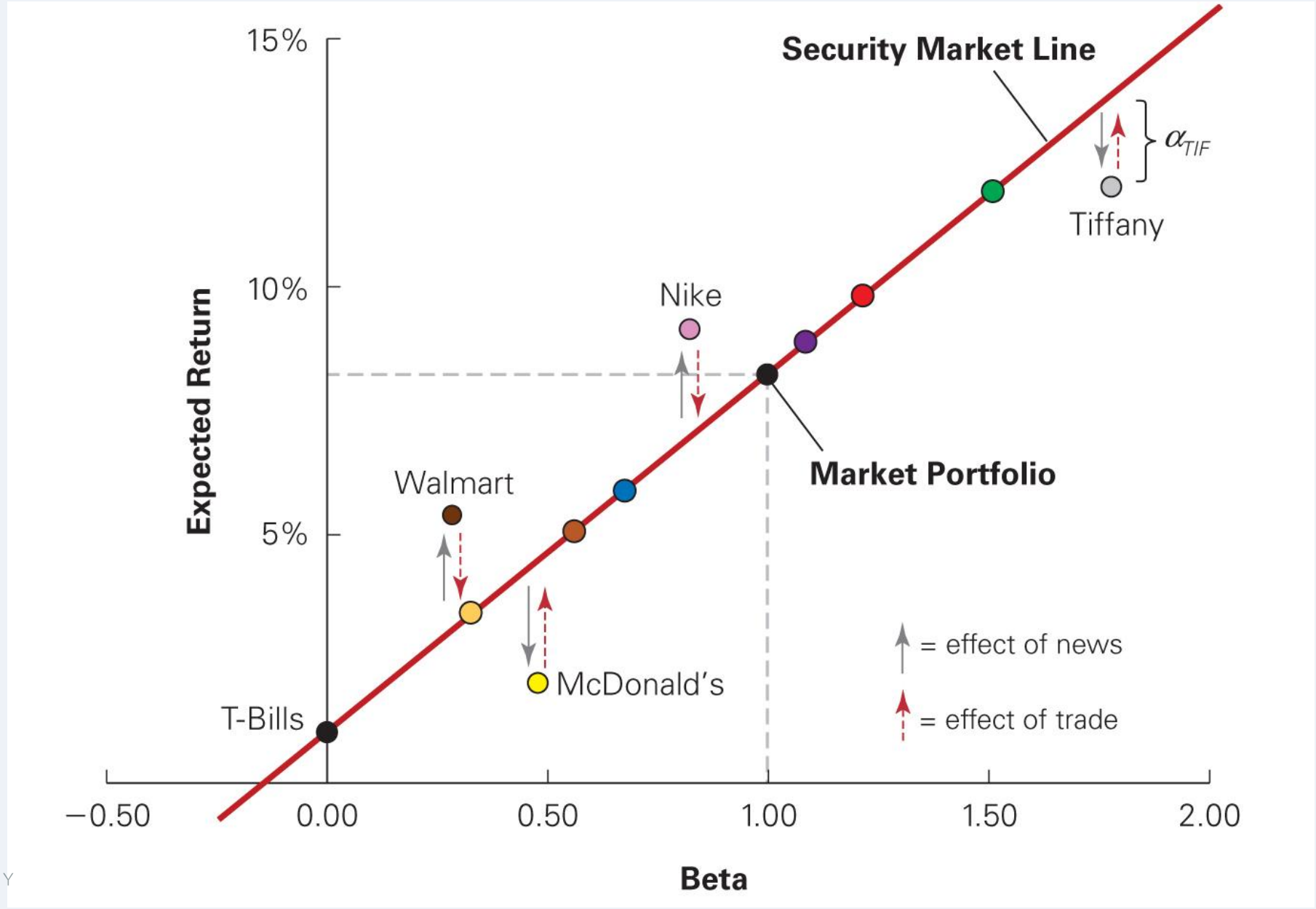
Using CAPM to find mispriced securities

- Because the SML is the graphical representation of the mean-beta relationship, "fairly priced" assets plot exactly on the SML.

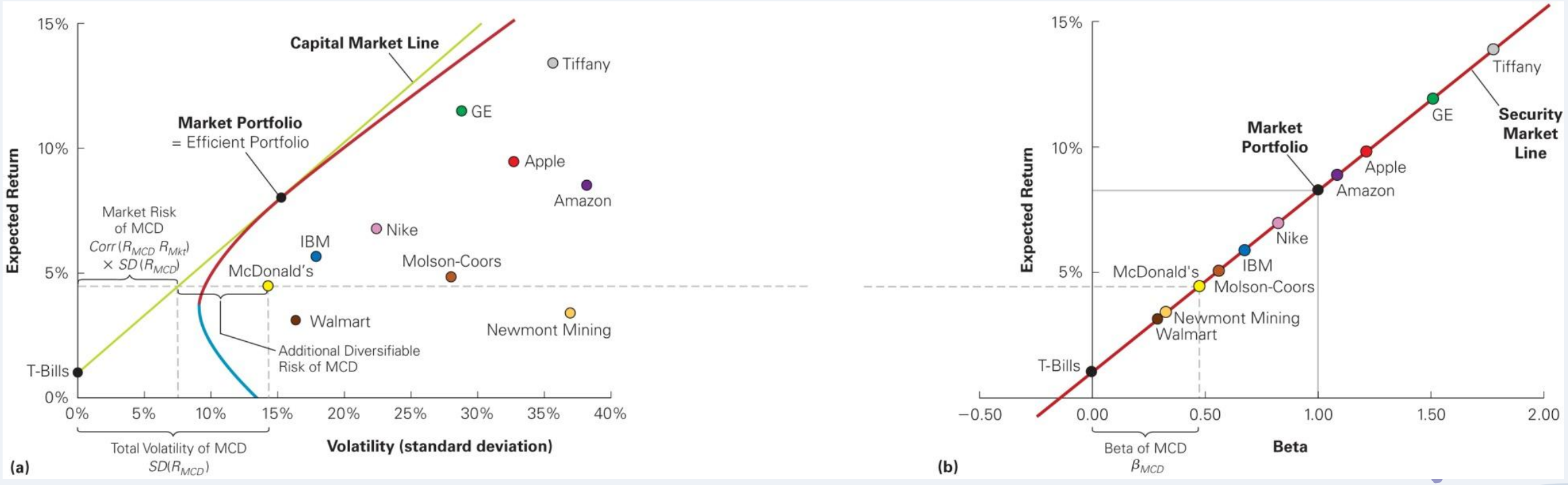
- For a given beta, if an asset is expected to provide higher return than the justified return required by the CAPM, we say the asset is earning some **positive abnormal return called Jensen's alpha** denoted by α_i . Assets with positive alpha are (underpriced / overpriced) and they plot (above / below) the SML.

- On the contrary, overpriced assets plot (above / below) the SML and they have (positive / negative) Jensen's alpha. They are expected to provide lower return than the CAPM's required return.

Jensen's Alpha



Differences between the CML and SML



Differences Between the CML and the SML

Features	SML	CML
1. Risk	Systematic risk (beta)	Standard deviation
2. Applicability	Can find the equilibrium return of any assets	Can find the equilibrium return of an efficient portfolio only
3. Slope	Market risk premium = $[\mathrm{E}(R_M) - r_f]$	Sharpe ratio = $\left[\frac{E(r_M)-r_f}{\sigma_M}\right]$

Single Index Model

- A single index model is a model that relates stock returns to returns on both a broad market index such as HSI and firm-specific factors.
- **Excess return** of an asset, R_i^e, is defined as the rate of return on the asset i in excess of the risk-free rate, (ri – rf).
- The index model can be represented by the following regression equation:

$$R_i^e = \alpha_i + \beta_i R_M^e + e_i. \quad (1)$$

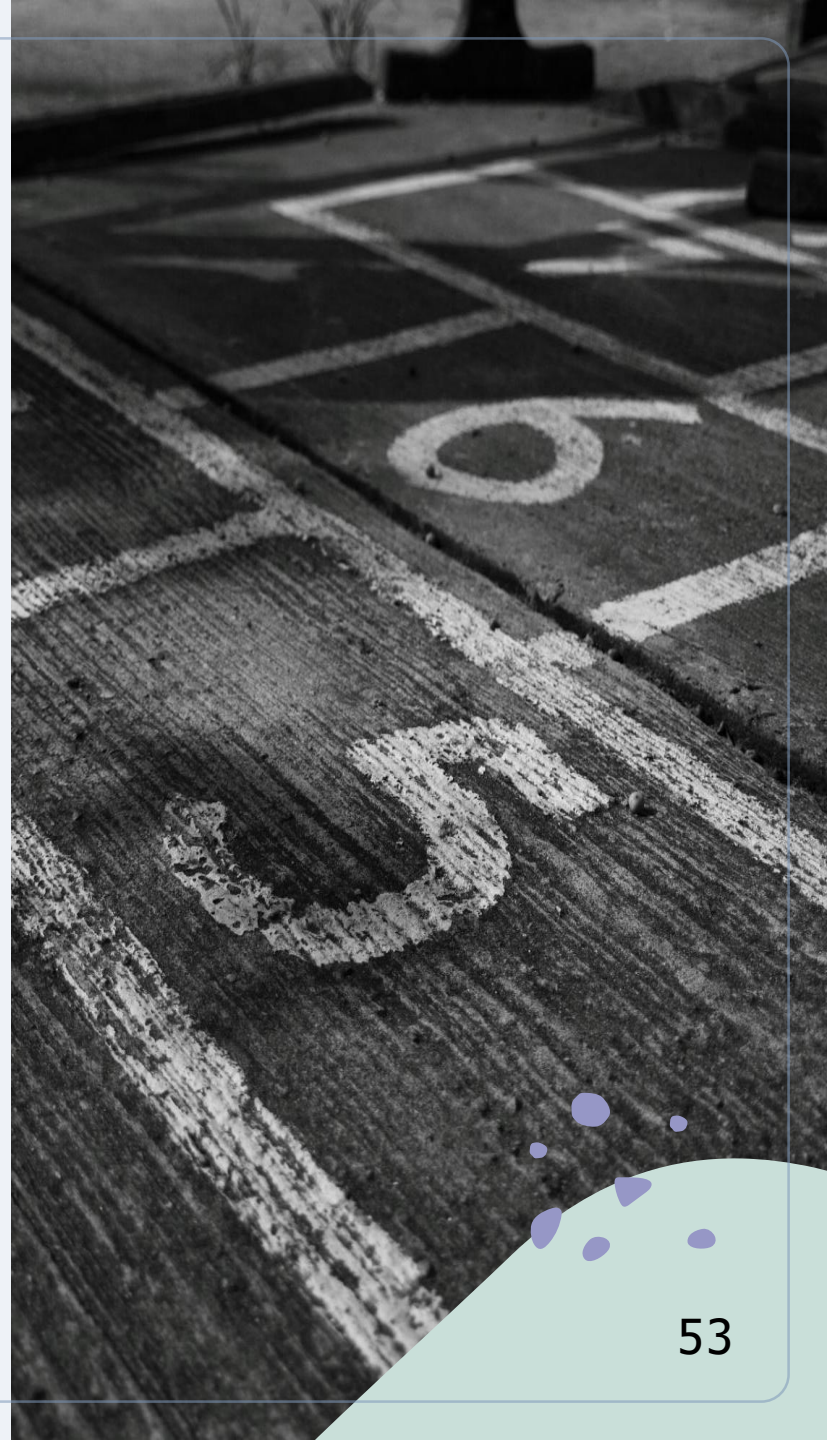
α_i: stock's expected excess return if market factor is neutral, that is, market-index excess return is zero.
R_M^e: the excess return on the market, r_M – rf.
β_i : the sensitivity of a security's return to the return on the market index
$\beta_i R_M^e$: component of return due to movements in overall market
e_i: Component attributable to unexpected events relevant only to this security (firm-specific).

Expected excess return of stock i

- If we take the expected value of Equation (1), the expectation of the firm-specific residual is zero and the **expected excess return of stock i** would be given by:

$$E(R_i^e) = \alpha_i + \beta_i E(R_M^e)$$

- We can estimate α_i and β_i by running a regression of the excess return on stock i against the excess return on the market index.

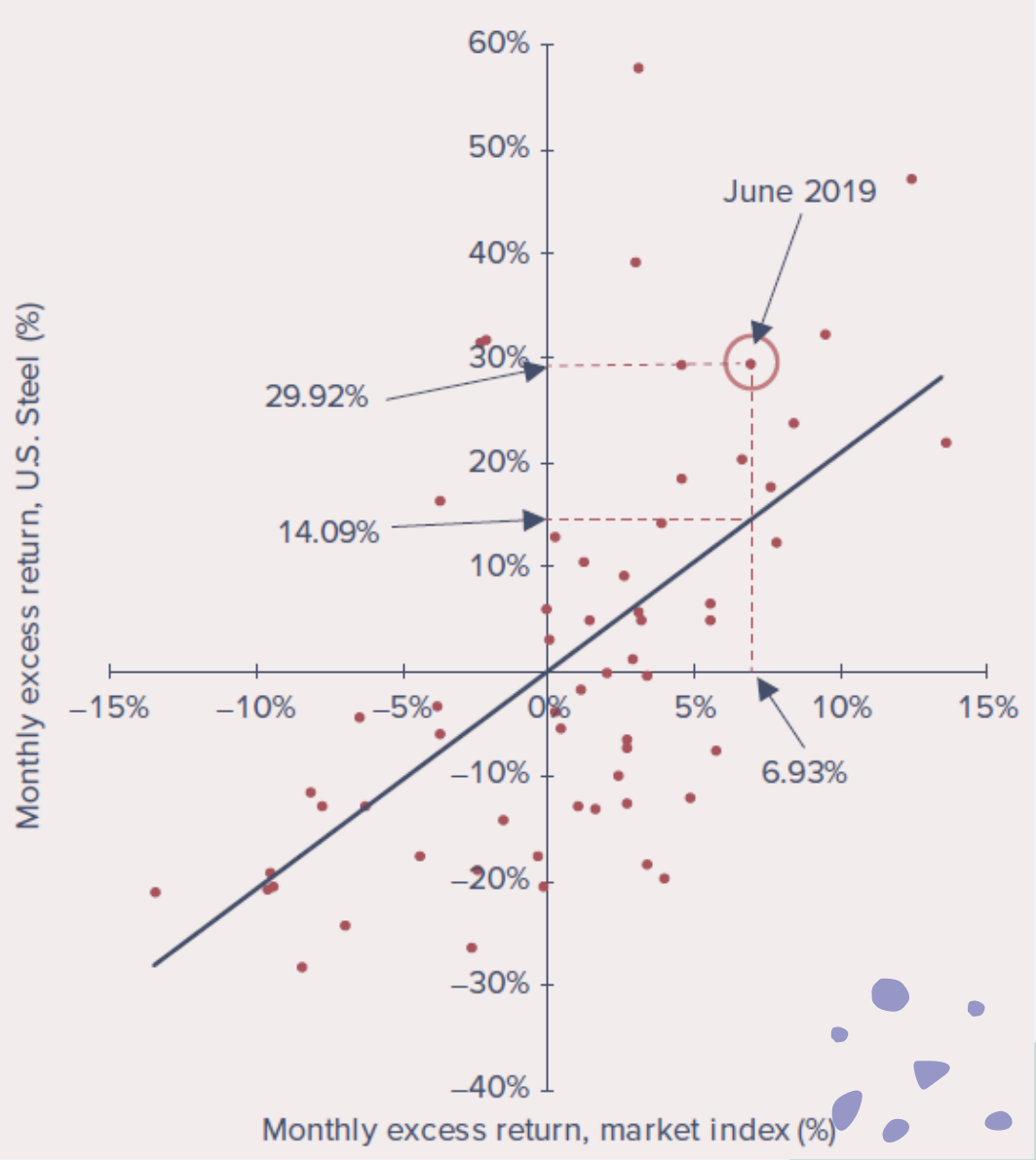


Scatter Diagram for U.S. Steel

- The regression line or best-fit line of the scatter plots is called **Security Characteristic Line (SCL)**. The algebraic representation of the regression line is:

$$E(R_i^e | R_M^e) = \alpha_i + \beta_i R_M^e$$

- **Interpretation:** The expected value of R_i^e given the value of R_M^e equals the intercept plus the slope coefficient times the value of R_M^e.



Meaning of beta of an asset, β_i

- Beta of the stock is just the slope of its Security Characteristic Line.
- Beta, β (a measure of systematic risk) estimates the sensitivity of the asset's rate of return to changes in the broad market's returns.

- It is the expected percent change in the excess return of the security for a 1% change in the excess return of the market portfolio. If beta is equal to 2, it means when the market return increases (decreases) by 1%, the return of the security will go up (down) by 2%.

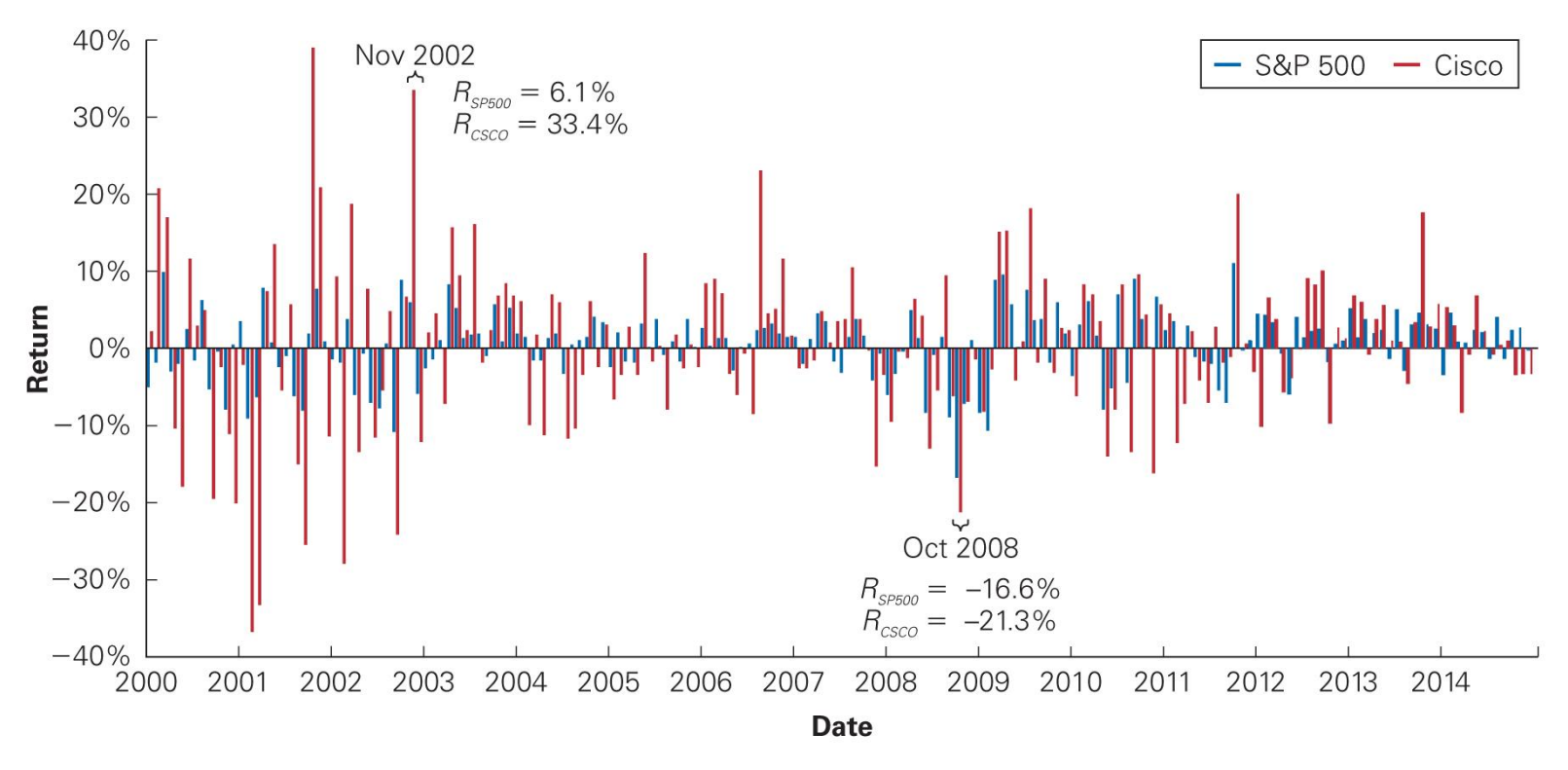
- In statistics, beta of a stock can be measured by, $\beta_i = \frac{Cov(r_i, r_M)}{\sigma_M^2}$

Meaning of beta of an asset, β_i

TABLE 6.6 Selected Beta Coefficients and Their Interpretations

Beta	Comment	Interpretation
2.0	Move in same direction as market	Twice as responsive as the market
1.0		Same response as the market
.5		Only half as responsive as the market
0		Unaffected by market movement
− .5	Move in opposite direction to market	Only half as responsive as the market
−1.0		Same response as the market
−2.0		Twice as responsive as the market

Monthly Returns for Cisco Stock and for the S&P 500, 2000-2015

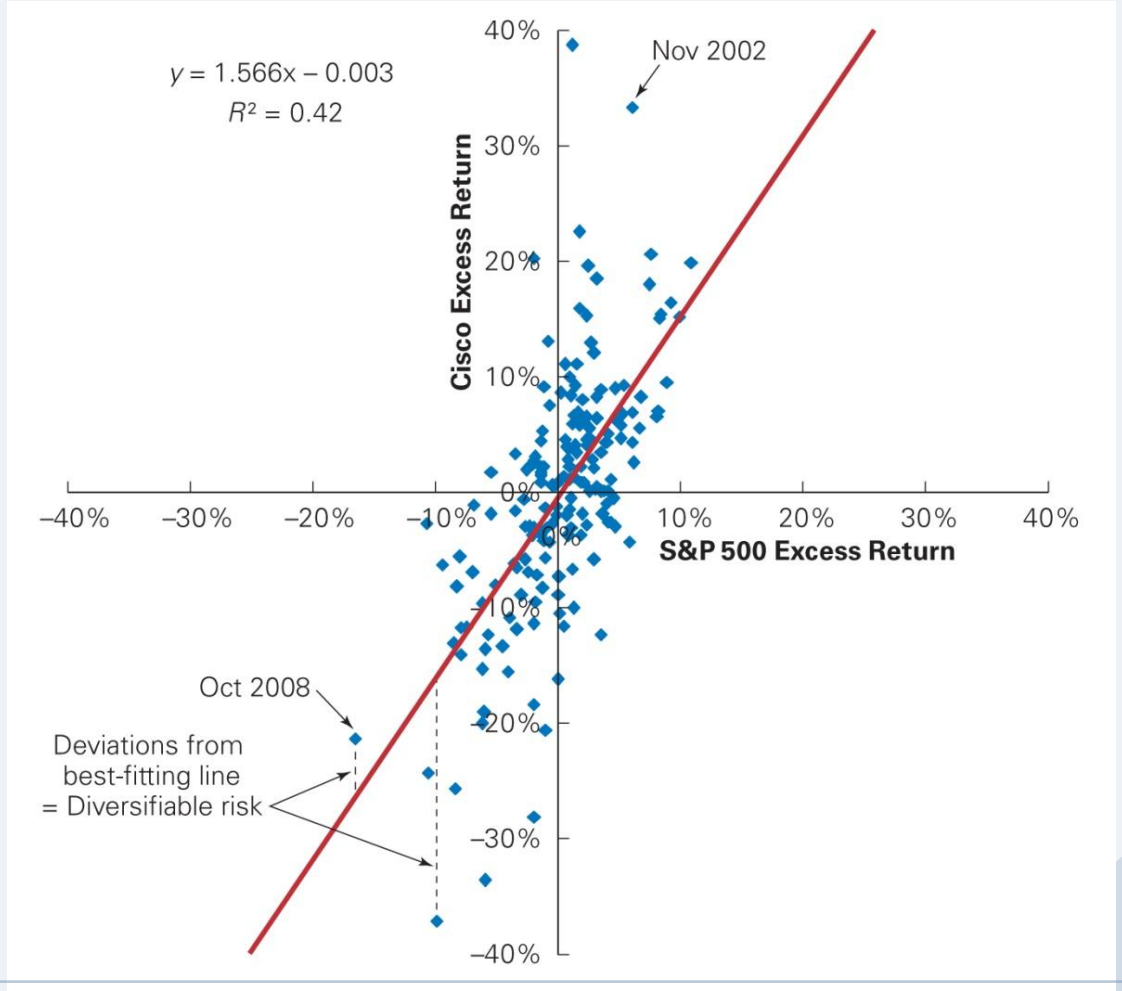


Scatterplot of Monthly Excess Returns for Cisco Versus the S&P 500, 2000-2015

If the regression line of Cisco is:

$$E(R^e_{Cisco}) = -0.003 + 1.566R^e_{S\&P\ 500}$$

1. Find the beta of Cisco.
2. Find the alpha of Cisco.



Betas with Respect to the S&P 500 for Individual Stocks (based on monthly data for 2010–2015)

Company	Ticker	Industry	Equity Beta
PG&E	PGE	Utilities	0.26
General Mills	GIS	Packaged Foods	0.30
Newmont Mining	NEM	Gold	0.32
The Hershey Company	HSY	Packaged Foods	0.33
McDonald's	MCD	Restaurants	0.39
Clorox	CLX	Household Products	0.40
Pepsico	PEP	Soft Drinks	0.42
Wal-Mart Stores	WMT	Superstores	0.49
Procter & Gamble	PG	Household Products	0.52
Coca-Cola	KO	Soft Drinks	0.52
Altria Group	MO	Tobacco	0.54
Amgen	AMGN	Biotechnology	0.64
Johnson & Johnson	JNJ	Pharmaceuticals	0.65
Nike	NKE	Footwear	0.67
Southwest Airlines	LUV	Airlines	0.80
Kroger	KR	Food Retail	0.80
Starbucks	SBUX	Restaurants	0.80
Whole Foods Market	WFM	Food Retail	0.83
Intel	INTC	Semiconductors	0.87
Microsoft	MSFT	Systems Software	0.89
Pfizer	PFE	Pharmaceuticals	0.89
Apple	AAPL	Computer Hardware	0.92
Amazon.com	AMZN	Internet Retail	0.94
Macy's	M	Department Stores	0.95
Foot Locker	FL	Apparel Retail	0.99

Betas with Respect to the S&P 500 for Individual Stocks (based on monthly data for 2010–2015)

Company	Ticker	Industry	Equity Beta
Alphabet (Google)	GOOG	Internet Software and Services	0.99
Molson Coors Brewing	TAP	Brewers	0.99
Harley-Davidson	HOG	Motorcycle Manufacturers	1.14
Yahoo!	YHOO	Internet Software and Services	1.21
salesforce.com	CRM	Application Software	1.22
Marriott International	MAR	Hotels and Resorts	1.24
Walt Disney	DIS	Movies and Entertainment	1.25
Coach	COH	Apparel and Luxury Goods	1.25
Cisco Systems	CSCO	Communications Equipment	1.27
Williams-Sonoma	WSM	Home Furnishing Retail	1.27
Staples	SPLS	Specialty Stores	1.36
Oracle	ORCL	Systems Software	1.42
Hewlett-Packard	HPQ	Computer Hardware	1.52
J. C. Penney	JCP	Department Stores	1.52
Wynn Resorts Ltd.	WYNN	Casinos and Gaming	1.59
Ryland Group	RYL	Homebuilding	1.60
Caterpillar	CAT	Construction Machinery	1.62
United States Steel	X	Steel	1.62
General Motors	GM	Automobile Manufacturers	1.66
Netgear	NTGR	Communications Equipment	1.91
Tiffany & Co.	TIF	Apparel and Luxury Goods	1.92
Autodesk	ADSK	Application Software	1.96
Ethan Allen Interiors	ETH	Home Furnishings	2.04
Advanced Micro Devices	AMD	Semiconductors	2.23
Sotheby's	BID	Auction Services	2.48

Source: CapitalIQ

Beta of an individual asset

The beta of an asset can be estimated by the following formula:

$$\beta_i = \frac{Cov_{i,M}}{\sigma_M^2} = \frac{\rho_{i,M}\sigma_i\sigma_M}{\sigma_M^2} = \rho_{i,M}\left(\frac{\sigma_i}{\sigma_M}\right)$$

where:

σ_i = standard deviation of returns for stock i

σ_M = standard deviation of returns for the market portfolio

$Cov_{i,M}$ = covariance between the returns for stock i and the market portfolio

$\rho_{i,M}$ = correlation between the returns for stock i and the market portfolio

σ_M^2 = variance of the returns on the market portfolio

Example

Assume the correlation of returns between Transport Co. and the market portfolio equals 0.80, the standard deviation of Transport Co. equals 0.60, and the standard deviation for the market portfolio equals 0.30. Calculate the beta for Transport Co.



Beta of a portfolio

- If we have estimated betas of assets, the beta of a portfolio is the weighted average beta of the securities in the portfolio.

$$\beta_P = \frac{Cov(R_P, R_{Mkt})}{Var(R_{Mkt})} = \frac{Cov\left(\sum_i x_i R_i, R_{Mkt}\right)}{Var(R_{Mkt})} = \sum_i x_i \frac{Cov(R_i, R_{Mkt})}{Var(R_{Mkt})} = \sum_i x_i \beta_i$$

Example

The Expected Return of a Portfolio

Problem

Suppose Kraft Foods' stock has a beta of 0.50, whereas Boeing's beta is 1.25. If the risk-free rate is 4%, and the expected return of the market portfolio is 10%, what is the expected return of an equally weighted portfolio of Kraft Foods and Boeing stocks, according to the CAPM?

Example

Solution

We can compute the expected return of the portfolio in two ways. First, we can use the SML to compute the expected return of Kraft Foods (KFT) and Boeing (BA) separately:

$$E[R_{KFT}] = r_f + \beta_{KFT}(E[R_{Mkt}] - r_f) = 4\% + 0.50(10\% - 4\%) = 7.0\%$$

$$E[R_{BA}] = r_f + \beta_{BA}(E[R_{Mkt}] - r_f) = 4\% + 1.25(10\% - 4\%) = 11.5\%$$

Then, the expected return of the equally weighted portfolio P is

$$E[R_P] = \tfrac{1}{2}E[R_{KFT}] + \tfrac{1}{2}E[R_{BA}] = \tfrac{1}{2}(7.0\%) + \tfrac{1}{2}(11.5\%) = 9.25\%$$

Alternatively, we can compute the beta of the portfolio using Eq. 11.24:

$$\beta_P = \tfrac{1}{2}\beta_{KFT} + \tfrac{1}{2}\beta_{BA} = \tfrac{1}{2}(0.50) + \tfrac{1}{2}(1.25) = 0.875$$

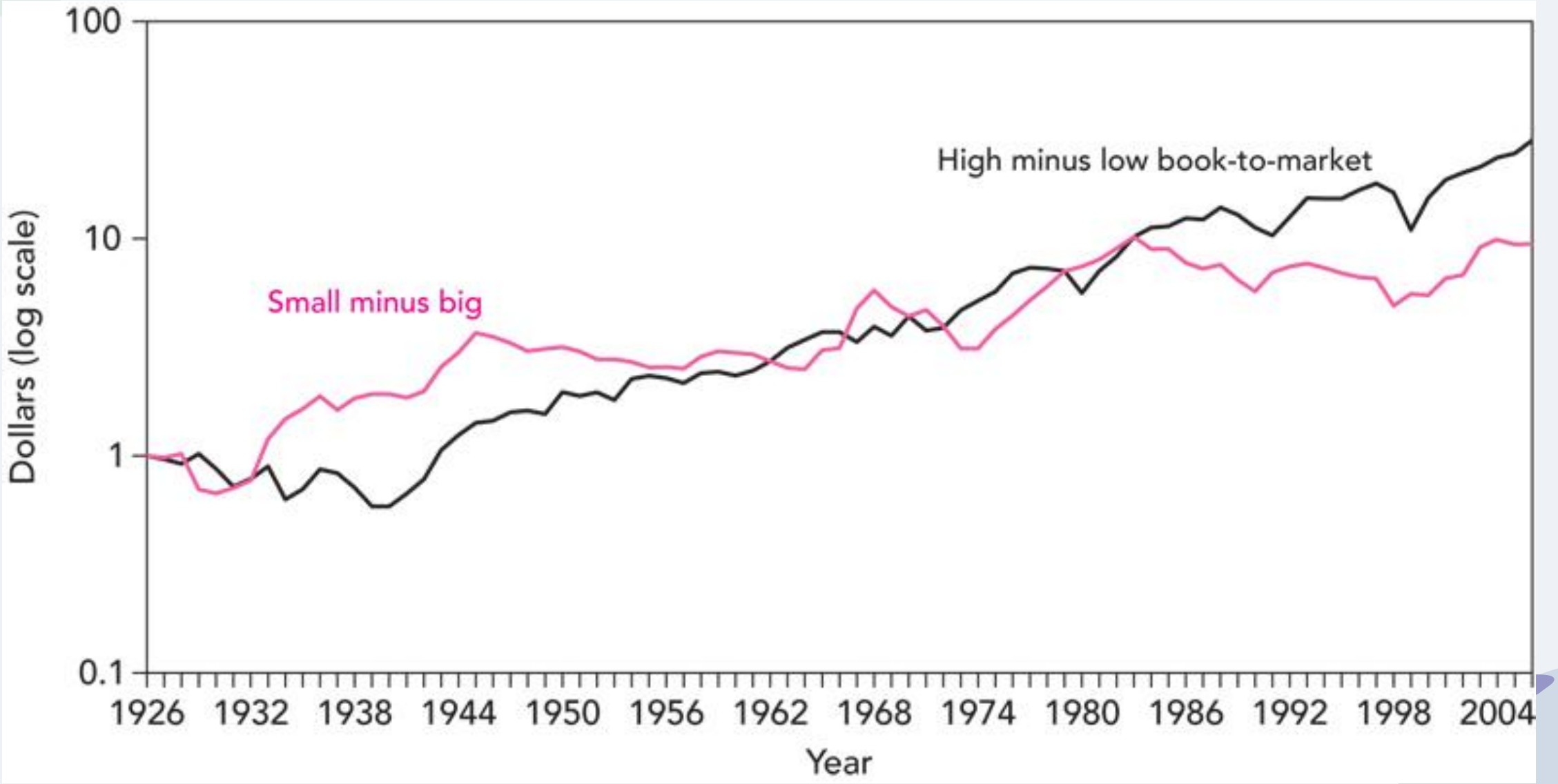
We can then find the portfolio's expected return from the SML:

$$E[R_P] = r_f + \beta_P(E[R_{Mkt}] - r_f) = 4\% + 0.875(10\% - 4\%) = 9.25\%$$

Shortcomings of the CAPM

- Lack of consideration of other factors: CAPM assumes that the only risk factor affecting an asset's return is the market risk, which may not always hold true. Other factors such as interest rate changes, political events, firm's size and book-to-market ratios can also impact asset returns. Since the end of 1926 the average annual difference between the returns on the small firms and larger firms has been 3.7%. Also, value stocks outperformed growth stocks by 5.2% for the same period.

Factors other than beta affect returns



Shortcomings of the CAPM

- Difficulty in estimating beta: Estimating an asset's beta can be challenging, as it requires historical data and assumptions about the relationship between the asset and the market portfolio.
- Difficulty in determining the composition of “true” market portfolio
- Too restrictive assumptions in the model: homogeneous expectations and normal distribution of returns.